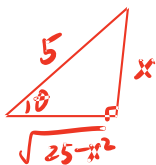


Evaluate the following integrals.

$$\begin{aligned}
 1). \int \frac{1}{x^2 \sqrt{25 - x^2}} dx & \quad \begin{aligned} x &= 5 \sin \theta \\ dx &= 5 \cos \theta d\theta \\ \sqrt{25 - x^2} &= \sqrt{25 - 25 \sin^2 \theta} \\ &= \sqrt{25(1 - \sin^2 \theta)} \\ &= \sqrt{25 \cos^2 \theta} = 5 \cos \theta \end{aligned} \\
 &= \int \frac{1}{25 \sin^2 \theta \cdot 5 \cos \theta} 5 \cos \theta d\theta = \int \frac{1}{25 \sin^2 \theta} d\theta \\
 &= \frac{1}{25} \int \frac{1}{\sin^2 \theta} d\theta \\
 &= \frac{1}{25} \int \csc^2 \theta d\theta \\
 &= \frac{1}{25} \cdot (-\cot \theta) + C \\
 &= -\frac{1}{25} \frac{\sqrt{25 - x^2}}{x} + C
 \end{aligned}$$

$$x = 5 \sin \theta \Rightarrow \sin \theta = \frac{x}{5} = \frac{O}{H}$$



$$\cot \theta = \frac{A}{O} = \frac{\sqrt{25 - x^2}}{x}$$

$$\begin{aligned}
 2). \int \frac{1}{x \sqrt{1 - x^2}} dx & \quad \begin{aligned} x &= \sin \theta \\ dx &= \cos \theta d\theta \\ \sqrt{1 - x^2} &= \sqrt{1 - \sin^2 \theta} \\ &= \sqrt{\cos^2 \theta} = \cos \theta \end{aligned} \\
 &= \int \frac{1}{\sin \theta \cos \theta} \cos \theta d\theta \\
 &= \int \frac{1}{\sin \theta} d\theta \\
 &= \int \csc \theta d\theta \\
 &= \ln | \csc \theta - \cot \theta | + C \\
 &= \ln \left| \frac{1}{x} - \frac{\sqrt{1 - x^2}}{x} \right| + C
 \end{aligned}$$

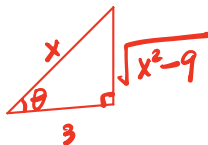
$$x = \sin \theta \Rightarrow \sin \theta = \frac{x}{1} = \frac{O}{H}$$



$$\begin{aligned}
 \csc \theta &= \frac{1}{\sin \theta} = \frac{1}{x} \\
 \cot \theta &= \frac{A}{O} = \frac{\sqrt{1 - x^2}}{x}
 \end{aligned}$$

$$\begin{aligned}
 & x = 3 \sec \theta \\
 3). \int \frac{\sqrt{x^2 - 9}}{x^3} dx & \quad dx = 3 \sec \theta \tan \theta d\theta \\
 & \quad \sqrt{x^2 - 9} = \sqrt{9 \sec^2 \theta - 9} = \sqrt{9(\sec^2 \theta - 1)} \\
 & \quad = \sqrt{9 \tan^2 \theta} = 3 \tan \theta \\
 & = \int \frac{3 \tan \theta \cdot 3 \sec \theta \tan \theta}{3^3 \sec^3 \theta} d\theta \\
 & = \frac{1}{3} \int \frac{\tan^2 \theta}{\sec^2 \theta} d\theta \\
 & = \frac{1}{3} \int \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos^2 \theta d\theta \\
 & = \frac{1}{3} \int \sin^2 \theta d\theta \\
 & = \frac{1}{3} \int \frac{1 - \cos 2\theta}{2} d\theta \\
 & = \frac{1}{6} \int (1 - \cos 2\theta) d\theta \\
 & = \frac{1}{6} \left(\theta - \frac{1}{2} \sin 2\theta \right) d\theta \\
 & = \frac{1}{6} \theta - \frac{1}{12} \cdot 2 \sin \theta \cos \theta + C \\
 & = \frac{1}{6} \theta - \frac{1}{6} \sin \theta \cos \theta + C \\
 & = \frac{1}{6} \operatorname{arccsc}\left(\frac{x}{3}\right) - \frac{1}{6} \cdot \frac{\sqrt{x^2 - 9}}{x} \cdot \frac{3}{x} + C \\
 & = \frac{1}{6} \operatorname{arccsc}\left(\frac{x}{3}\right) - \frac{1}{2} \cdot \frac{\sqrt{x^2 - 9}}{x^2} + C
 \end{aligned}$$

$$x = 3 \sec \theta \Rightarrow \frac{x}{3} = \sec \theta = \frac{H}{A}$$



$$\theta = \operatorname{arccsc}\left(\frac{x}{3}\right)$$

$$\sin \theta = \frac{\sqrt{x^2 - 9}}{x}$$

$$\cos \theta = \frac{3}{x}$$

$$x = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$

$$\sqrt{x^2 + 1} = \sqrt{\tan^2 \theta + 1} = \sec \theta$$

$$\text{When } x=0: \tan \theta = 0 \Rightarrow \theta = 0$$

$$x=1: \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$\begin{aligned}
 4). \int_0^1 \sqrt{x^2 + 1} dx & \\
 & = \int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta
 \end{aligned}$$

use formula on formula sheet

$$= \left[\frac{1}{2} \sec \theta \cdot \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{1}{2} \sec\left(\frac{\pi}{4}\right) \tan\left(\frac{\pi}{4}\right) + \frac{1}{2} \ln \left| \sec\left(\frac{\pi}{4}\right) + \tan\left(\frac{\pi}{4}\right) \right| \right)$$

$$- \left(\frac{1}{2} \sec 0 \tan 0 + \frac{1}{2} \ln |\sec 0 + \tan 0| \right)$$

$$= \frac{1}{2} \cdot \sqrt{2} \cdot 1 + \frac{1}{2} \ln(\sqrt{2} + 1) - \left(\frac{1}{2} \cdot 0 + \frac{1}{2} \ln(1 + 0) \right)$$

$$= \frac{\sqrt{2}}{2} + \frac{1}{2} \ln(\sqrt{2} + 1)$$

$$5). \int_1^2 \frac{\sqrt{x^2-1}}{x} dx$$

$$x = \sec \theta$$

$$dx = \sec \theta \cdot \tan \theta d\theta$$

$$\sqrt{x^2-1} = \sqrt{\sec^2 \theta - 1} = \tan \theta$$

$$\text{When } x=1, \sec \theta = 1 \Rightarrow \theta = 0$$

$$\text{When } x=2, \sec \theta = 2 \Rightarrow \theta = \frac{\pi}{3}$$

$$= \int_0^{\frac{\pi}{3}} \frac{\tan \theta}{\sec \theta} \cdot \sec \theta \cdot \tan \theta d\theta$$

$$= \int_0^{\frac{\pi}{3}} \tan^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{3}} (\sec^2 \theta - 1) d\theta$$

$$= (\tan \theta - \theta) \Big|_0^{\frac{\pi}{3}}$$

$$= \left(\tan\left(\frac{\pi}{3}\right) - \frac{\pi}{3} \right) - (\tan 0 - 0)$$

$$= \sqrt{3} - \frac{\pi}{3}$$

$$6). \int \frac{1}{x\sqrt{9x^2+1}} dx$$

$$x = \frac{1}{3} \tan \theta$$

$$dx = \frac{1}{3} \sec^2 \theta d\theta$$

$$\sqrt{9x^2+1} = \sqrt{9 \cdot \left(\frac{1}{3} \tan \theta\right)^2 + 1}$$

$$= \sqrt{9 \cdot \frac{1}{9} \tan^2 \theta + 1}$$

$$= \sqrt{\tan^2 \theta + 1}$$

$$= \sec \theta$$

$$= \int \frac{1}{\frac{1}{3} \tan \theta \sec \theta} \cdot \frac{1}{3} \sec^2 \theta d\theta$$

$$= \int \frac{\sec \theta}{\tan \theta} d\theta$$

$$= \int \frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} d\theta$$

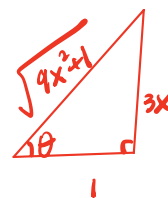
$$= \int \frac{1}{\sin \theta} d\theta$$

$$= \int \csc \theta d\theta$$

$$= \ln |\csc \theta - \cot \theta| + C$$

$$= \ln \left| \frac{\sqrt{9x^2+1}}{3x} - \frac{1}{3x} \right| + C$$

$$x = \frac{1}{3} \tan \theta \Rightarrow \tan \theta = 3x = \frac{3x}{1} = \frac{O}{A}$$



$$\csc \theta = \frac{H}{O} = \frac{\sqrt{9x^2+1}}{3x}$$

$$\cot \theta = \frac{A}{O} = \frac{1}{3x}$$

$$9 - 25x^2 = 25\left(\frac{9}{25} - x^2\right)$$

$$\Rightarrow x = \frac{3}{5} \sin \theta \quad \sqrt{9 - 25x^2} = \sqrt{9 - 25 \cdot \left(\frac{9}{25} \sin^2 \theta\right)}$$

$$dx = \frac{3}{5} \cos \theta d\theta \quad = \sqrt{9 - 9 \sin^2 \theta}$$

$$7). \int_0^{0.6} \frac{x^2}{\sqrt{9 - 25x^2}} dx \quad = 3 \cos \theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{3}{5} \sin \theta\right)^2}{3 \cos \theta} \cdot \frac{3}{5} \cos \theta d\theta \quad \text{When } x=0: \frac{3}{5} \sin \theta = 0 \Rightarrow \theta = 0$$

$$\text{When } x=0.6: \frac{3}{5} \sin \theta = 0.6$$

$$\Rightarrow \sin \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$= \frac{9}{25} \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta$$

$$= \frac{9}{125} \int_0^{\frac{\pi}{2}} \frac{1 - \cos(2\theta)}{2} d\theta$$

$$= \frac{9}{125} \left(\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right) \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{9}{125} \left[\left(\frac{1}{2} \cdot \frac{\pi}{2} - \frac{1}{4} \sin 0 \right) - \left(\frac{1}{2} \cdot 0 - \frac{1}{4} \sin 0 \right) \right]$$

$$= \frac{9\pi}{500}$$

$$u = \sin t$$

$$du = \cos t dt$$

$$\text{When } t=0: u=0$$

$$t = \frac{\pi}{2}: u = \sin\left(\frac{\pi}{2}\right) = 1$$

$$8). \int_0^{\pi/2} \frac{\cos t}{\sqrt{1 + \sin^2 t}} dx$$

$$= \int_0^1 \frac{1}{1+u^2} du$$

$$= \arctan u \Big|_0^1$$

$$= \arctan 1 - \arctan 0$$

$$= \frac{\pi}{4}$$